LECTURE 1: BASICS

· PIMPS – players, info, moves, payoffs, strategies

· Static - one, simult. Shot game

· Dynamic – sequential game

· Common Knowledge: if everyone know about E and everyone knows that everyone knows about E

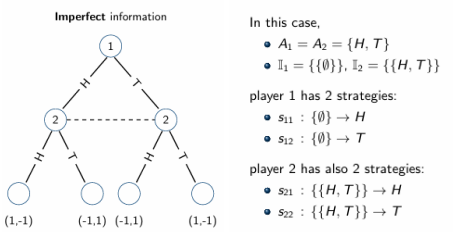
· Complete Information: if the following are common knowledge to all players: all potential actions, outcomes, how each combination of actions affect the outcomes, and the preference of every plater over outcomes

LECTURE 2: NORMAL FORM GAMES

· Elements of extensive form game: tree, actions, root / internal or decision / leaf or terminal node, terminal history (a path in the tree)

· Information set (I): set of their own decision nodes (and of their history) that are indistinguishable to them

· Strategy (si): of player i is a mapping si: I1 → Ai, such that si(J) ∈ C(J) , ∀J ∈ Ii



· Perfect Information: players can observe actions and outcomes from opponents. *In this example, P2 would have 4 strategies.

· Best response strategy: argmax u(s,s-i), always chosen if rational player.

LECTURES 3&4: DOMINANCE & NASH

· Strictly dominant strategy: if player i is rational, regardless of the strategy chosen by the other players, this strategy will always yield the maximum payoff. (IESDS)

· Nash Equilibrium: for a normal form game, a pure strategy NE is the one where not a single player could be better off by unilaterally deviating from their strategy. It is the concentration of best responses.

How to identify NE? For each of P2's strategy, what is the best strategy I (P1) should use.

· Pareto dominant NE: if it improves the pay-off of at least 1 player, without making others worse off.

· Risk dominant NE: the one that minimizes the product of the loss exposure of all players, if they unilaterally deviate from their equilibrium strategy.

LECTURE 5: MIXED STRATEGIES

· Definition: For a player i with a set Si of pure strategies, a mixed strategy σi is a probability distribution over Si in range [0,1].

· Expected payoff: combination of pay-offs from application of the opponents' pure strategies, weighted by their corresponding probability. The support set of a mixed strategy only includes non-zero prob. strats.

· Nash Existence Theorem (Nobel Prize in Economics in 1994): for any normal form game with finite number of players and strategy sets, there exists at least one mixed-strategy Nash equilibrium.

LECTURE 6: BAYESIAN GAMES

Must mention (1) set N={1,2} of players, (2) types of players theta1 = c and theta2=(a,b), (3) beliefs of players such as p(c/a)=p(c/b)=1 and p(a/c)=p and p(b/c)=1-p, (4) actions of players (X,Y) and (5) payoffs in matrix

· Selten Game Equivalent: converts a game of incomplete information into a game of complete information by adding players, i.e 1 for each players type. *Mention nº of players and expected payoffs in table.

· Mixed Strategies: σ1 = (σ1(X),σ1(Y), where σ1(X) = p and σ1(Y) = 1−p

σ2 = (σ2(X),σ2(Y)); σ2(X) = q and σ2(Y) = 1−q

Find p by: u2(σ1,X) = u2(σ1,Y)

Solution: (σ1*, σ2*)=((p,1-p),(q,1-q))

*If asked for best response, provide function depending of values of "p".

LECTURES 7&8: MECHANISM DESIGN

· Social Choice Function: Considering a set of agents N = {1,...,n}, with sets of types Θ1,...,Θn. Given a set of outcomes X, a social choice function is a mapping of the form f : Θ1 × ..., ×Θn → X. It is said to be implementable if no players have an interest in lying to obtain a higher payoff

· Direct Mechanism: Given a social choice function f : Θ1 × ..., ×Θn → X, a direct mechanism is defined as the tuple ⟨Θ1,...,Θn,f ⟩.

· Indirect Mechanism: instead of relying on types, and indirect mechanism considers the connection between the strategies used by the agents, given their type, and the outcomes; defined as a tuple ⟨S1,...,Sn,g⟩.

· Incentive-compatible: mechanism that is implementable in some type of equilibrium, where no agents involved has an incentive not to report, or act upon, their true type

· Open auction: all agents can openly observe the process and its evolution until a winner is revealed (English and Dutch) – dynamic games or incomplete information

· Closed auction: all agents communicate in a private and secretive way with the mechanism designer. Winner determined by secretly communicated info. Static games of incomplete information (first/second-price sealed bid auction)

In second-price sealed auctions, betting your valuation is a weakly dominant strategy.

$$u_i(b_i, b_{-i}, \theta_i) = \begin{cases} \theta_i - b_j^*, & \text{if } b_i > b_j,\ \forall j \neq i, \text{ and } b_j^* = max_{j \neq i}\ b_j \\ 0, & \text{if } \exists j:\ b_i \leq b_j \end{cases}$$

If player 1 increases their bid over θ1, their pay-off would become negative. Player 1 would never do that. If lowering their bid too much, it could go below θ2, and then the pay-off would be 0. Player 1 would also not do that. And, in general, if placing bids lower than θ1, that would decrease their pay-off. Therefore, there is no strategy that achieves a higher pay-off than b1 = θ1.

In a first-price sealed-bid auction, it a strictly dominated strategy to place a bid bi that is equal to their valuation θi, or that is greater than their valuation (i.e., bi ≥ θi).

LECTURES 9: CORRELATION/CONTRACTS

· Security Value (v): In a normal form game, the minimum guaranteed pay-off if freely choosing any strategy si ∈ Si as a response to the strategy of others (s−i ∈ S−i), i.e vi = maxsi∈Si mins−i∈S−i ui(si,s−i).

How to identify? For each of my strategies, collect the values for the worst P2 can do to you. Then, select the maximum out of them.

· Individual Rationality: A correlated strategy τC ∈ ΔSC is said to be individually rational for all players i ∈ C if the expected pay-offs from the correlated strategy are greater than their security values, i.e., ui(τC) ≥ vi, ∀i ∈ C.

· Contract: is one type of interaction among players. It comprises an agreement prior to, or during the game from a strategic point of view, we consider that players may sign it or not. If they sign it, they will respect it.

Prisoner's Dilemma with AC1 (NC,NC):

	NC	C	AC1	AC2
NC	$-4,-4$	$-6,0$	$-6,0$	$-6,0$
C	$0,-6$	$-5,-5$	$-5,-5$	$-5,-5$
AC1	$0,-6$	$-5,-5$	$-4,-4$	$-5,-5$
AC2	$0,-6$	$-5,-5$	$-5,-5$	$-3,-3$

For (NC,AC1), P2 will apply security strategy of that row (C) if they know P1 will not AC1.

For AC2, they decide to toss a coin; if heads implement (NC,C), if tails (C,NC).

If only P1 wants to AC2 and knows P2 won't (AC2,NC), then P1 will act as if they applied C (replace payoffs in C,NC in AC2,NC)

· Correlated Strategy: probability distribution over the set of potential combinations of their individual strategies. Sc = { (NC,NC),(NC,C),(C,C),(C,NC)} – as a group, you can increase expected payoffs

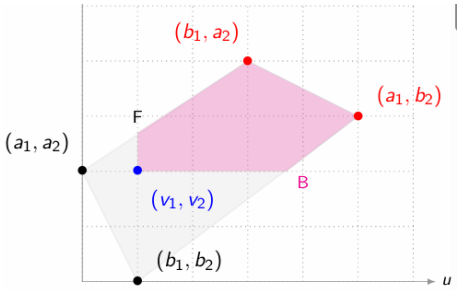
LECTURE 10: BARGAINING

· Nash program (why cooperate?): Any normal form game Γ = ⟨N,(Si),(ui)⟩ can be transformed into a more cooperative game, based on interaction, by extending the strategy sets of all players and allowing for correlated strategies and potentially higher expected payoffs for the agents.

· Feasibility Set: all possible expected payoffs that can be achieved by applying any correlated strategy τ (a probability distribution over all pure strategy profiles). It forms a convex hull, representing all outcomes that are attainable in principle through combinations of pure strategies.

· Disagreement Point: strategy profile that is necessarily Pareto-dominated, defining the minimum payoffs players should get, whatever happens. Defined by security vals.

· Reduced feasibility set: section of the original feasibility set, reduced by the disagreement point.



Solution: Nash Bargaining solution proposes an optimisation problem that somewhat maximises the pay-offs for both players.

$$(u_1^*, u_2^*) = \underset{u_1,u_2}{\operatorname{argmax}} \quad (u_1 - v_1)(u_2 - v_2) \quad \text{s.t.} \quad (u_1, u_2) \in F, \quad u_1 \geq v_1, \; u_2 \geq v_2$$

*We multiply instead of adding them to ensure "fairness", i.e if one received zero it ensures that the other one has zero too.

NASH AXIOMS

· Strong Pareto efficiency: cannot find another point on F where players would get a strictly higher payoff

· Symmetry: if F and the disagreement point (v1=v2) are symmetric, then the solution of Nash bargaining problem is symmetric too.

· Independence of irrelevant alternatives: if there is other bargaining problem that is within (F,v), they will have same solution.

· Scale invariance: if we apply a linear transformation to (F,v) such that (G,w), then the solution follows the same transformation.

OTHER SOLUTIONS

$$(u_1^*, u_2^*) = \underset{u_1,u_2}{\operatorname{argmax}} \quad u_1 + u_2 \quad \text{s.t.} \quad u_1 - v_1 = u_2 - v_2, \quad (u_1, u_2) \in F, \quad u_1 \geq v_1, \; u_2 \geq v_2$$

This is Egalitarian. Utilitarian removes the constraint highlighted in red.

LECTURE 11: COALITIONAL GAMES

2^n number of coalitions formed, N players

Characteristic function are the payoffs

Imputation:

$x_i \geq v(\{i\})$ (individual rationality)
$\sum_i x_i = v(N)$ (collective rationality)

Domination:

$\sum_{i \in \mathcal{C}} x_i \leq v(\mathcal{C})$, and
$x_i \geq x_i', \; \forall i \in \mathcal{C}$. (with strict inequality

Monotonicity: if when enlarging a coalition, its value increases

$$\mathcal{C} \subseteq \mathcal{D} \implies v(\mathcal{C}) \leq v(\mathcal{D}), \; \forall \mathcal{C} \subseteq \mathcal{D} \subseteq N$$

Super-additivity: when the value of the coalition is greater than the sum of its individual components

$$v(\mathcal{C} \cup \mathcal{D}) \geq v(\mathcal{C}) + v(\mathcal{D}), \; \forall \mathcal{C}, \mathcal{D} \subseteq N,$$

Convexity: if the benefit of adding an additional player to a given coalition gets higher for larger coalitions, i.e the grand coalition is most worth it for all players

$$v(\mathcal{C} \cup \mathcal{D}) + v(\mathcal{C} \cap \mathcal{D}) \geq v(\mathcal{C}) + v(\mathcal{D}),$$

Inessential game: when there is no benefit in joining any coalitions. Only one imputation. If not, it's an essential game.

$$v(N) = \sum_{i \in N} v(\{i\})$$

LECTURE 12: THE CORE

Coalitional Rationality: an allocation (x,y,z) if it each players' individual payoffs are greater than their split in other given coalitions.

$$\sum_{i \in \mathcal{C}} x_i \geq v(\mathcal{C}), \; \forall \mathcal{C} \subseteq N$$

The core: is linked to Nash Equilibrium; for all allocations inside it, players have no interest in unilaterally deviating from them. They must be individually, collectively and coalitionally rational. The core is convex and compact. It can be empty.

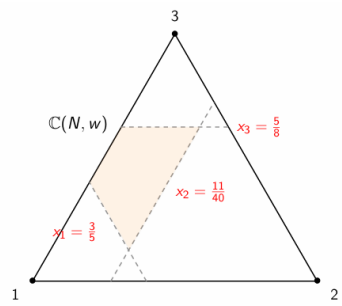
Visualisation = normalise the game

Individual payoffs = 0; Grand coalition = 1

Pairs of n-1 become:

$$w(\mathcal{C}) = \frac{1}{v(N) - \sum_{i \in N} v(\{i\})} \left(v(C) - \sum_{i \in \mathcal{C}} v(\{i\}) \right)$$

Equilateral triangle (h=1). Each vertex defines allocation (1,0,0), (0,1,0), (0,0,1)



Find boundaries by applying coalitional rationality constraints as follows:

X1+x2 >= 0.3 , therefore x3 < 0.7

*Can also solve with LP given

$$x^* = \underset{x}{\operatorname{argmin}} \quad x_1 + \ldots + x_n \quad \text{s.t.} \quad \sum_{i \in \mathcal{C}} x_i \geq v(\mathcal{C}), \; \forall \mathcal{C} \subseteq N, \quad x_1, \ldots, x_n \geq 0$$

LECTURE 13: SHAPLEY VALUE

Efficiency axiom: states that, to be fair to all players, all of the joint pay-offs should be allocated (formula on left side)

Dummy player: one for which pay-offs do not increase when they join any coalition. It is only fair that they get zero in allocations.

$$\sum_{i \in N} \psi_i(v) = v(N) \qquad v(\mathcal{C} \cup \{i\}) = v(\mathcal{C})$$

Interchangable player: symmetry axiom state that it is only fair that these players receive the same split.

Shapley Value: pay-off to be given to player i ∈ N for their average marginal contribution over all possible coalitions.

$$\psi_i = \frac{1}{n} \sum_{\mathcal{C} \subseteq N \setminus i} \binom{n-1}{|\mathcal{C}|}^{-1} \delta(i, \mathcal{C} \cup i)$$

If the game is convex, the shapley value is in the core. If not, it is not guaranteed. Then, we need to check for individual and coalitional rationality (collective is already ensured).

LECTURE 14: STABLE MATCHING

Perfect Matching: the association of every element of O with every element of D.

Problem? Unstable pair for stablished preferences. No pair of elements should feel they could have jointly been better off. No guarantee of stability.

$$o_i, \; d_j \succ \phi(o_i)$$
$$d_j, \; o_i \succ \phi^{-1}(d_j)$$

Gale-Shapley Algorithm:

- Always converges to a stable allocation (perfect and stable matching).

- It is bounded as it requires at most n^2 proposals. For the O-version, it is O-optimal and D-pessimal; and vice versa.

- It is also incentive-compatible for O, but not for D. This implies that it is a dominant strategy for O to express their true preference, and not a dominant strategy for D to express theirs